DISTRIBUTED SOVEREIGN AI
INFRASTRUCTURE NODES TECHNOLOGY

SOVEREIGN INFRASTRUCTURE | DISTRIBUTED BY DESIGN | SUSTAINABLE VALUE | COMMUNITY POWERED

BUSINESS MODEL

BUILDING GERMANY'S SOVEREIGN AI INFRASTRUCTURE BACKBONE

INFRASTRUCTURE OWNERSHIP
We own and operate the infrastructure – not the GPUs.

SUSTAINABLE BY DESIGN
Energy-efficient, heat-reuse and renewable-powered sites.

STRONG PARTNERSHIPS
Collaboration with municipalities, industry and public institutions.

LOW RISK, HIGH IMPACT
Resilient business model with long-term value creation.

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DSAINT

DISTRIBUTED SOVEREIGN AI INFRASTRUCTURE NODES TECHNOLOGY

BUSINESS MODEL OVERVIEW

Updated Strategic & Technical Edition

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DSAINT develops and operates distributed sovereign AI infrastructure as an Infrastructure-as-a-Platform (IaaP) business, with AI workload orchestration and trusted federation as the technology layer connecting physical and distributed compute.

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This overview updates the established DSAINT business model in line with the current prototype-first, team-based development strategy.

1. Executive Business Model Summary

DSAINT is structured as an Infrastructure-as-a-Platform (IaaP) company focused on the development, ownership and operation of distributed sovereign AI infrastructure nodes across Germany and, ultimately, Europe.

The business model combines physical infrastructure, AI compute enablement, secure federation, workload orchestration, operations and ecosystem services. DSAINT's long-term advantage is intended to come from connecting professionally operated infrastructure with research, enterprise and future community compute resources through a common trusted orchestration layer.

The updated strategy changes the sequence of value creation. Rather than requiring DSAINT to finance and construct a national infrastructure network immediately, the first commercial and technical asset is the DSAINT orchestration/federation capability. Physical Node 01 then becomes the reference infrastructure platform through which the model can be validated and scaled.

Business sequence: technology validation → team/partner integration → reference infrastructure → recurring infrastructure services → distributed federation scale.

2. Business Model Architecture

Layer	DSAINT role	Value created
Infrastructure ownership	Own/develop strategic physical infrastructure or coordinate ownership structures	Long-term control of sovereign infrastructure assets
Platform operations	Operate AI-ready infrastructure and associated facilities	Reliable, secure and efficient infrastructure services
Orchestration platform	Develop and operate workload orchestration and federation software	Efficient placement and management of distributed AI workloads
Federation	Connect trusted external compute resources	Additional capacity, resilience and geographic reach
Ecosystem	Coordinate municipalities, enterprises, research and technology partners	Demand, expertise, innovation and deployment opportunities
Community model	Future Reverse Co-Hosting / Community Compute Federation	Broader participation and distributed capacity, subject to validation

3. Customer and User Segments

Segment	Need	DSAINT proposition
AI companies / software companies	Reliable AI compute	Sovereign, managed AI infrastructure and workload services
German & European enterprises	Trusted AI infrastructure	Secure capacity without requiring every customer to build its own AI facility
Research institutions / universities	Flexible compute and experimentation	Federated access to appropriate resources
Municipalities / public-sector organizations	Regional digital sovereignty	Local infrastructure participation and access to sovereign AI capabilities

Industrial users	AI workloads near relevant data and operations	Trusted regional compute and orchestration
Infrastructure partners	Efficient utilization of facilities	Federation and orchestration layer
Future community participants	Monetize or contribute available compute	Controlled Reverse Co-Hosting model

4. Value Proposition

- Sovereign: infrastructure designed around German/European control and trusted operation.
- Distributed: capacity can be geographically distributed rather than concentrated in a single facility.
- Federated: heterogeneous resources can participate through a common orchestration model.
- Operational: customers access infrastructure services rather than managing all underlying complexity themselves.
- Secure: identity, access, workload and infrastructure security are integrated into the platform.
- Energy-aware: high-density AI infrastructure is designed around efficient cooling, power and potential heat recovery.
- Scalable: Node 01 provides a reference architecture that can be repeated across additional locations.

5. Products and Services

Offering	Description	Maturity
AI Infrastructure Platform	Managed sovereign AI-ready infrastructure capacity	Long-term / Node 01
AI Workload Orchestration	Scheduling, placement, monitoring and policy management across heterogeneous resources	Immediate technical focus
Federated Compute Services	Access to distributed trusted compute resources	Prototype / future
Infrastructure-as-a-Service	Compute, storage, networking and related infrastructure services	Future commercial service
Managed AI Infrastructure	Operational support for customers requiring dedicated environments	Future
Node Development Services	Reference designs and deployment support for future DSAINT nodes	Future
Federation Membership / Participation	Participation by approved organizations and compute providers	Future
Community Compute Services	Reverse Co-Hosting and contribution accounting	Research / future

6. Revenue Model

The exact pricing model is not yet fixed. The business model should initially validate demand and unit economics rather than assume final tariffs.

Revenue stream	Potential mechanism	Priority
Infrastructure capacity	Recurring fee for reserved or consumed capacity	High
Managed infrastructure	Recurring operations/service fee	High
AI workload orchestration	Platform/service fee associated with	High

	orchestration	
Federated compute	Usage-based or capacity-based service fee	Medium
Dedicated environments	Long-term contracts for enterprise/research customers	High
Node participation services	Technical onboarding and operational services	Medium
Future federation contribution accounting	Service/coordination fee where legally and economically appropriate	Long term
Heat / energy integration	Potential ancillary value from recovered heat	Long term / site dependent

7. Ownership Model

The established DSAINT ownership principle separates ownership and control of infrastructure from ownership of individual compute hardware. This allows DSAINT to build a durable infrastructure business without requiring the company to own every GPU participating in the ecosystem.

Asset / role	Possible DSAINT position
Land	Own, lease or operate through a project/site structure
Buildings	Own or control through long-term contractual arrangements
Power infrastructure	Own/control or operate through infrastructure partnerships
Cooling infrastructure	Own/control or operate
Physical security	Own/control operational responsibility
Connectivity	Multi-carrier contracts and network partnerships
GPU/compute hardware	May be DSAINT-owned, customer-owned or partner-owned depending on deployment
Orchestration software	DSAINT-controlled technology/platform
Federation governance	DSAINT-led with appropriate stakeholder governance

Core principle: DSAINT can own and operate the infrastructure platform without needing to own every compute resource in the federation.

8. Operating Model

- DSAINT develops the reference architecture and platform.
- DSAINT operates or coordinates strategic infrastructure nodes.
- Customers consume infrastructure and platform services.
- Technology partners provide specialized systems and components.
- Research and enterprise partners contribute use cases and validation.
- External compute providers can participate under defined trust and operational policies.
- Node 01 becomes the first integrated operational environment.

8.1 Core Operational Capabilities

Capability	Function
Infrastructure Operations	Power, cooling, facilities, physical security and maintenance
NOC	Network, compute and service monitoring
SOC	Cybersecurity monitoring and incident response
Platform Operations	Workload orchestration, federation and service management
Customer Operations	Onboarding, support, SLA management and reporting
Partner Management	Technical, research, municipal and commercial relationships
Compliance	Security, privacy, procurement and regulatory controls

9. Cost Structure

Cost category	Examples
Site / infrastructure CAPEX	Land, building, electrical, cooling, security, connectivity infrastructure
Compute CAPEX	GPUs, servers and related hardware where DSAINT owns capacity
Software development	Orchestration, federation, monitoring and security platform
Personnel	Engineering, infrastructure operations, security, sales and management
Energy	Electricity, cooling and associated utility costs
Connectivity	Carrier and network services
Maintenance	Hardware, facilities and technical support
Compliance / insurance	Security, legal, regulatory and insurance costs
Customer acquisition	Sales, partnerships and ecosystem development

The detailed financial model should be maintained separately. This overview intentionally avoids unsupported CAPEX, OPEX, pricing or margin assumptions.

10. Business Development Stages

Stage	Business objective	Primary evidence
1 – Technical foundation	Develop DSAINT expertise and prototype capability	Working laboratory
2 – Federation prototype	Demonstrate orchestration and distributed execution	Multi-node prototype and benchmarks
3 – Team integration	Join/collaborate with experienced organizations	Technical partnerships / team role
4 – Commercial validation	Identify anchor users and use cases	Letters of interest, pilot workloads, contracts
5 – Node 01	Establish reference physical platform	Commissioned demonstrator
6 – Recurring services	Generate stable infrastructure/platform revenue	Customer contracts and utilization
7 – Network expansion	Replicate validated node model	Additional nodes
8 – Federation scale	Connect external resources	Growing trusted federation

11. Competitive Positioning

The current strategy recognizes that distributed AI infrastructure, sovereign cloud, HPC and AI orchestration are already active fields in Germany and Europe. DSAINT should therefore avoid positioning itself simply as 'another distributed AI network'.

The business proposition should instead be tested around the combination of:

- Physical sovereign infrastructure ownership/operation
- AI workload orchestration across heterogeneous nodes
- Trusted federation of independently operated resources
- Regional and municipal infrastructure participation
- Future Reverse Co-Hosting / Community Compute Federation
- Energy-aware infrastructure operation
- Repeatable node architecture

Differentiation must be demonstrated through technical performance, operational advantages, customer demand and federation economics — not only through a conceptual description.

12. Partner Ecosystem

Partner type	Role
Municipalities	Sites, regional development, infrastructure and ecosystem support
Data-center engineering firms	Design and implementation
Energy/grid partners	Power supply, grid integration and energy optimization
Cooling specialists	Liquid cooling and thermal systems
Network providers	Multi-carrier connectivity
AI/GPU infrastructure companies	Compute hardware and integration
Research institutions	Algorithms, orchestration, benchmarking and validation
AI companies	Anchor workloads and demand
Cybersecurity organizations	Security architecture and assurance
Investors / financing partners	Future infrastructure expansion
Technology teams	Co-development and engineering expertise

13. Updated Ecosystem Strategy

Following the recent discussion with the German federal administration, DSAINT's practical development model is explicitly team-based. The project should seek to become technically useful within the existing ecosystem rather than assume that one founder can independently develop and deploy the full infrastructure stack.

1. Map companies and research groups working on adjacent problems.
2. Identify gaps where DSAINT can contribute technically.
3. Build the workload-orchestration prototype as evidence of capability.
4. Join or collaborate with experienced infrastructure/AI teams.
5. Use partner projects to validate architecture and operating assumptions.
6. Develop DSAINT's independent infrastructure business in parallel as the evidence base grows.

14. Community Compute and Reverse Co-Hosting

The long-term business concept includes Citizen AI Nodes and Reverse Co-Hosting. Participants could contribute compute capacity and receive compensation based on measured contribution.

This mechanism is not treated as a mature revenue stream at the current stage. It requires validation of security, workload isolation, energy costs, accounting, taxation/legal structure, liability, reliability and economic incentives.

Requirement	Validation question
Security	Can external compute safely execute approved workloads?
Measurement	Can contribution be measured objectively?
Economics	Is contribution worth more than the participant's incremental cost?
Reliability	Can workload availability be maintained despite node churn?
Governance	Who defines participation and workload policies?
Compliance	How are data protection, liability and contractual obligations handled?

15. Unit Economics Framework

Before scaling Node 01 or the wider network, DSAINT should measure the economics of one unit of useful AI compute.

Metric	Purpose
Revenue per compute unit	Measure customer value
Energy cost per compute unit	Measure direct operating cost
Cooling cost per compute unit	Measure thermal overhead
Network cost per workload	Measure distributed execution overhead
Infrastructure depreciation per compute unit	Understand capital intensity
Operations cost per compute unit	Measure staffing and service burden
Utilization	Determine revenue-generating capacity
Gross margin	Evaluate commercial viability

The business model succeeds only if sovereign AI infrastructure can deliver sufficient customer value at sustainable utilization and operating economics.

16. Business Risks and Mitigations

Risk	Mitigation
Capital intensity	Stage investment; separate prototype from infrastructure financing.
Competition	Focus on measurable differentiation and ecosystem integration.
Insufficient utilization	Develop anchor customers and research workloads before expansion.
Founder capacity	Build a multidisciplinary team and partner network.

Technology change	Modular architecture and hardware/vendor flexibility.
Energy prices	Efficiency, energy strategy and workload-aware scheduling.
Security liability	Security-by-design and strong operational controls.
Federation complexity	Start with trusted partners before community-scale participation.
Regulatory uncertainty	Dedicated legal/compliance workstream.

17. Business KPIs

Area	Indicative KPI
Demand	Qualified customer pipeline / anchor workloads
Infrastructure	Utilization of available AI capacity
Revenue	Recurring revenue and revenue per deployed MW
Economics	Gross margin and operating cost per compute unit
Platform	Successful workload scheduling and execution rate
Federation	Number and quality of trusted participating nodes
Reliability	Service availability
Security	Incidents and response performance
Sustainability	PUE, energy intensity and recovered heat
Growth	Number of validated nodes / deployments

18. Updated Business Roadmap

Phase	Indicative period	Business focus
Foundation	2026–2027	Skills, prototype, ecosystem mapping and technical partnerships
Validation	2027–2028	Federation prototype, workloads, benchmarks and first commercial use cases
Reference node	2027+	Node 01 feasibility, financing and physical implementation when gates are met
Early operations	2028–2030	Managed infrastructure and orchestration services
German expansion	2030–2034	Additional Core Infrastructure Nodes based on validated economics
Federation expansion	2030+	Research, enterprise and controlled community participation
European expansion	2034–2036+	Cross-border sovereign federation where technically and legally feasible

Dates are strategic targets rather than fixed commitments. Physical deployment depends on feasibility, partners, financing, permitting and evidence.

19. Organizational Model

Function	Required capability
Founder / Strategy	Vision, partnerships, business development and coordination
Technical Leadership	Distributed systems, AI infrastructure and architecture
Platform Engineering	Orchestration, APIs, scheduling and federation
Infrastructure Engineering	Power, cooling, networking and data-center systems
Security	Cybersecurity, identity, compliance and incident response
Operations	NOC/SOC, facilities and service management
Commercial	Customers, contracts and ecosystem development
Finance / Legal	Project finance, contracts, ownership and compliance

The organization should grow in stages. Not all functions need to be internal from the beginning; strategic partners can provide specialized capabilities.

20. Funding Strategy

DSAINT's funding strategy should be staged according to maturity.

Stage	Potential funding logic
Prototype	Founder resources, collaboration, research/innovation support where available
Technical validation	Research partnerships, grants and strategic technology collaboration
Node 01 feasibility	Strategic partners, project development capital and public/private financing mechanisms
Node 01 construction	Project finance, infrastructure capital, strategic investors and contracted demand
Network expansion	Project-level financing and retained operating cash flow

Public funding opportunities may support appropriate research or infrastructure components, but the business model should not depend on a single public funding source.

21. Long-Term Business Model

At maturity, DSAINT is intended to operate as a distributed sovereign AI infrastructure platform with multiple revenue-generating nodes and a trusted federation layer.

- DSAINT-owned or controlled strategic infrastructure nodes
- Recurring enterprise and research infrastructure services
- AI workload orchestration platform services
- Federated compute services
- Managed infrastructure and operations
- Technical ecosystem and node-development services
- Future community compute coordination services

Long-term model: infrastructure assets + software platform + recurring services + trusted

federation.

22. Conclusion

The updated DSAINT Business Model Overview preserves the project's core IaaP concept while making the route to commercial viability more realistic.

The immediate business asset is not a nationwide data-center network. It is the combination of technical capability, orchestration technology, partner relationships and validated demand that can later support physical infrastructure investment.

Node 01 remains central: it is the reference physical platform through which DSAINT can demonstrate sovereign infrastructure, workload orchestration, federation operations, security and sustainability. If the technical and economic assumptions are validated, the model can be replicated across additional nodes.

DSAINT: build the technology, validate the economics, operate the infrastructure, and federate the capacity.

23. Document Status

Item	Information
Project	DSAINT – Distributed Sovereign AI Infrastructure Nodes Technology
Business model	Infrastructure-as-a-Platform (IaaP)
Founder	Ernest Repko
Location	Germany
Email	info@dsaint.eu
Website	www.dsaint.eu
Document	Business Model Overview – Updated Strategic & Technical Edition
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Status	Strategic business model overview; detailed financial assumptions maintained separately